

Tarea

2.23) a) $\text{CH}_4 = \text{C} = 12.011$
 $\text{H} = 1.00794 \times 4 = 4.03$

$$\text{CH}_4 = 16.04 \text{ uma}$$

b) $\text{NO}_2 = \text{N} = 14.0067$
 $\text{O} = 15.9994 \times 2 = 32 = 46.01 \text{ uma}$

c) $\text{SO}_3 = \text{S} = 32.066$
 $\text{O} = 15.9994 \times 3 = 48 = 80.07 \text{ uma}$

d) $\text{C}_6\text{H}_6 = \text{C} = 12.011 \times 6 = 72.07 = 78.12 \text{ uma}$
 $\text{H} = 1.00794 \times 6 = 6.05$

e) $\text{NaI} = \text{Na} = 22.98977 = 149.90 \text{ uma}$
 $\text{I} = 126.9045$

f) $\text{K}_2\text{SO}_4 = \text{K} = 39.0983 \times 2 = 78.1966$
 $\text{S} = 32.066$
 $\text{O} = 15.9994 \times 4 = 64 = 164.26 \text{ uma}$

g) $\text{Ca}_3(\text{PO}_4)_2 = \text{Ca} = 40.078 \times 3 = 120.234$
 $\text{P} = 30.97376$
 $\text{O} = 15.9994 \times 8 = 128 = 248.21 \text{ uma}$

$$2.24) a) \text{Li}_2\text{CO}_3 = \text{Li} = 6.941 \times 2 = 13.90 \\ \text{C} = 12.011 \\ \text{O} = 15.9994 \times 3 = 48 \\ = 74 \text{ g/mol}$$

$$b) \text{CS}_2 = \text{C} = 12.011 \\ \text{S} = 32.066 \times 2 = 64.132 \\ = 76.14 \text{ g/mol}$$

$$c) \text{CHCl}_3 = \text{C} = 12.011 \\ \text{H} = 1.00794 \\ \text{Cl} = 35.4527 \times 3 = 106.3581 \\ = 119.47 \text{ g/mol}$$

$$d) \text{C}_6\text{H}_8\text{O}_6 = \text{C} = 12.011 \times 6 = 72.066 \\ \text{H} = 1.00794 \times 8 = 8.06352 \\ \text{O} = 15.9994 \times 6 = 95.9964 \\ = 176.13 \text{ g/mol}$$

$$e) \text{KNO}_3 = \text{K} = 39.0983 \\ \text{N} = 14.0067 \\ \text{O} = 15.9994 \times 3 = 47.9982 \\ = 101.10 \text{ g/mol}$$

$$f) \text{Mg}_3\text{N}_2 = \text{Mg} = 24.305 \times 3 = 72.915 \\ \text{N} = 14.0067 \times 2 = 28.134 \\ = 101.05 \text{ g/mol}$$

$$3.25) 152 \text{ g/mol} \times \frac{1 \text{ g}}{0.372 \text{ mol}} = 408.60 \text{ g}$$

$$152 \text{ g} \times \frac{1 \text{ mol}}{408.60 \text{ g}} = 0.37 \text{ mol}$$

$$3.26) \text{C}_2\text{H}_6 = 30.07 \text{ g}$$

$$0.334 \text{ g C}_2\text{H}_6 \times \frac{1 \text{ mol}}{30.07 \text{ g C}_2\text{H}_6} \times \frac{1 \text{ mol}}{1 \text{ mol}}$$

$$3.27) \text{C}_4\text{H}_{12}\text{O}_6 = 180.16 \text{ g}$$

$$1.50 \text{ g C}_4\text{H}_{12}\text{O}_6 \times \frac{1 \text{ mol}}{180.16 \text{ g}} \times \frac{1 \text{ mol C}}{1 \text{ mol C}_4\text{H}_{12}\text{O}_6} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol C}} = 3 \times 10^{22} \text{ atoms C}$$

$$1.50 \text{ g C}_4\text{H}_{12}\text{O}_6 \times \frac{1 \text{ mol}}{180.16 \text{ g}} \times \frac{12 \text{ mol H}}{1 \text{ mol C}_4\text{H}_{12}\text{O}_6} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol H}} = 6.02 \times 10^{22} \text{ atoms H}$$

$$1.50 \text{ g C}_4\text{H}_{12}\text{O}_6 \times \frac{1 \text{ mol}}{180.16 \text{ g}} \times \frac{6 \text{ mol O}}{1 \text{ mol C}_4\text{H}_{12}\text{O}_6} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol O}} = 3 \times 10^{22} \text{ atoms O}$$

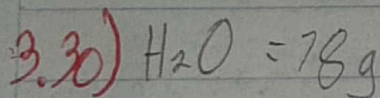
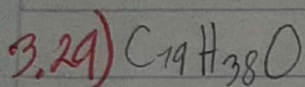
$$3.28) [(\text{CH}_3)_2\text{SO}] = 63.37 \text{ g}$$

$$7.74 \times 10^3 \text{ g} \times \frac{1 \text{ mol}}{63.37 \text{ g}} \times \frac{2 \text{ mol C}}{1 \text{ mol } (\text{CH}_3)_2\text{SO}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol C}} = 1.36 \times 10^{26} \text{ atoms C}$$

$$7.14 \times 10^3 \text{ g} \times \frac{1 \text{ mol}}{63.3 \text{ g}} \times \frac{1 \text{ mol S}}{1 \text{ mol } (\text{C}_2\text{H}_2)_2\text{S}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol S}} = 6.79 \times 10^{25} \text{ atoms}$$

$$7.14 \times 10^3 \text{ g} \times \frac{1 \text{ mol}}{63.3 \text{ g}} \times \frac{3 \text{ mol H}}{1 \text{ mol } (\text{C}_2\text{H}_2)_2\text{S}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol H}} = 2.08 \times 10^{26} \text{ atoms}$$

$$7.14 \times 10^3 \text{ g} \times \frac{1 \text{ mol}}{63.3 \text{ g}} \times \frac{1 \text{ mol O}}{1 \text{ mol } (\text{C}_2\text{H}_2)_2\text{S}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol O}} = 6.79 \times 10^{25} \text{ atoms}$$



$$2.56 \text{ mol H}_2\text{O} \times \frac{1 \text{ g H}_2\text{O}}{1 \text{ mol H}_2\text{O}} \times \frac{1 \text{ mol H}_2\text{O}}{18 \text{ g H}_2\text{O}} \times \frac{6.022 \times 10^{23} \text{ molecules}}{1 \text{ mol H}_2\text{O}} = 8.56 \times 10^{22} \text{ molecules}$$

3.31) Mide razones masa/carga de iones, calibrando un haz de material del compuesto a analizar hasta vaporizarlo e ionizar los diferentes átomos.

3.32) se puede medir o determinar mediante una técnica llamada espectrometría de masas.

3.33)

CF 4

126c
120c

15c
6c

= 98.9%

3.34)

34s
16s

36s
6s

= 104.3%

3.34)

$$\frac{34\text{ g}}{16\text{ g}}$$

$$\frac{36\text{ g}}{4\text{ g}}$$

$$= 104.3\%$$

3.35) La composición porcentual es una medida de la cantidad de masa que ocupa un elemento en un compuesto. Se mide en porcentaje de masa.

3.36) Para poder identificarlo es un poco más fácil con la fórmula: $\frac{\text{peso atómico} \times \text{número de átomos en la molécula}}{\text{peso molecular}} \times 100$

3.37) Es una expresión que representa los átomos que forman un compuesto químico sin atender a su estructura.

3.38) Se necesita el peso molecular y el número de átomos.

3.39) SnO_2 =

$$\begin{aligned}\text{Sn} &= 118.710 \div 150.7088 = 0.80 \times 100 = 80\% \text{ Sn} \\ \text{O} &= 15.9994 \times 2 = 31.9988 \div 150.7088 = 0.20 \times 100 = 20\% \text{ O} \\ &\quad 150.7088 \text{ g/mol}\end{aligned}$$

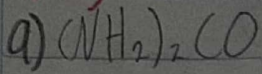
3.40) CHCl_3

$$\begin{aligned}\text{C} &= 12.011 \div 119.37704 = 0.10 \times 100 = 10\% \text{ C} \\ \text{H} &= 1.00794 \div 119.37704 = 8.44 \times 10^{-3} \times 100 = 0.84\% \text{ H} \\ \text{Cl} &= 35.4527 \times 3 = 106.3581 \div 119.37704 = 0.89 \times 100 = 89\% \text{ Cl} \\ &\quad 119.37704\end{aligned}$$

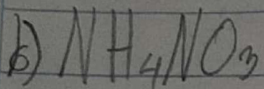
3.41) $\text{C}_9\text{H}_{10}\text{O}$

$$\begin{aligned}\text{C} &= 12.011 \times 9 = 108.099 \div 134.1778 = 0.81 \times 100 = 81\% \text{ C} \\ \text{H} &= 1.00794 \times 10 = 10.0794 \div 134.1778 = 0.08 \times 100 = 8\% \text{ H} \\ \text{O} &= 15.9994 \div 134.1778 = 0.12 \times 100 = 12\% \text{ O} \\ &\quad 134.1778\end{aligned}$$

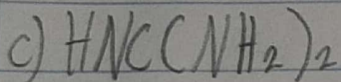
3.42)



$$\begin{aligned} \text{N} &= 14.0067 \div 44.03298 = 0.32 \times 100 = 32\% \text{ N} \\ \text{H} &= 1.00794 \times 2 = 2.01588 \div 44.03298 = 0.05 \times 100 = 5\% \\ \text{C} &= 12.011 \div 44.03298 = 0.27 \times 100 = 27\% \\ \text{O} &= 15.9994 \div 44.03298 = 0.36 \times 100 = 36\% \\ &\quad \underline{44.03298} \end{aligned}$$



$$\begin{aligned} \text{N} &= 14.0067 \times 2 = 28.01 \div 80.01614 = 0.35 \times 100 = 35\% \text{ N} \\ \text{H} &= 1.00794 \times 4 = 4.00794 \div 80.01614 = 0.05 \times 100 = 5\% \\ \text{O} &= 15.9994 \times 3 = 47.9982 \div 80.01614 = 0.60 \times 100 = 60\% \\ &\quad \underline{80.01614} \end{aligned}$$



$$\begin{aligned} \text{H} &= 1.00794 \times 3 = 3.00794 \div 43.03 = 0.07 \times 100 = 7\% \\ \text{N} &= 14.0067 \times 2 = 28.01 \div 43.03 = 0.65 \times 100 = 65\% \text{ N} \\ \text{C} &= 12.011 \div 43.03 = 0.28 \times 100 = 28\% \\ &\quad \underline{43.03} \end{aligned}$$

d) $\text{NH}_3 \rightarrow$ Es la que tiene más porcentaje de nitrógeno

$$\begin{aligned} \text{N} &= 14.0067 \div 17.01 = 0.82 \times 100 = 82\% \text{ N} \\ \text{H} &= 1.00794 \times 3 = 3.00794 \div 17.01 = 0.18 \times 100 = 18\% \\ &\quad \underline{17.01} \end{aligned}$$

3.43)

$$C \ 44.4\% = 44.4g \cdot \frac{1 \text{ mol}}{12.011g} = 3.70$$

6

$$H \ 6.21\% = 6.21g \cdot \frac{1 \text{ mol}}{1.00794g} = 6.16$$

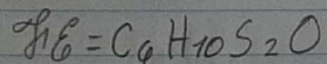
70

$$S \ 39.5\% = 39.5g \cdot \frac{1 \text{ mol}}{32.066g} = 1.23$$

2

$$O \ 9.86\% = 9.86g \cdot \frac{1 \text{ mol}}{15.9994g} = 0.62$$

1



3.44)

$$C = 19.8\% = 19.8g \cdot \frac{1 \text{ mol}}{12.011g} = 1.65$$

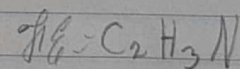
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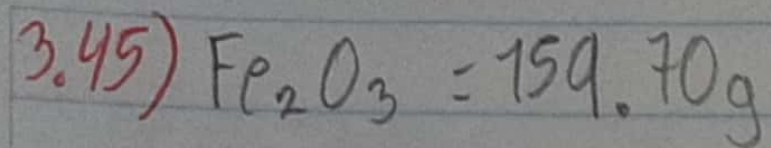
$$H = 2.50\% = 2.50g \cdot \frac{1 \text{ mol}}{1.00794g} = 2.48$$

3

$$N = 11.6\% = 11.6g \cdot \frac{1 \text{ mol}}{14.0067g} = 0.83$$

1





$$24.6\text{g} \times \frac{1\text{mol}}{159.70\text{g}} = 0.15\text{mol Fe}$$